Reza Akbari Movahed

PhD Student in Computing Science

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ABOUT ME

Motivated Data Scientist and Machine Learning Engineer specialising in deep learning for biomedical image and signal analysis. Currently a PhD student in Computing Science at the University of Glasgow, researching cardiac motion estimation from cine CMR using Bayesian deep learning, 3D mesh modelling, and generative AI. Previous research includes AI-based diagnosis of depression and mild cognitive impairment, sperm segmentation, and medical image watermarking.

EXPERIENCE

Graduate Teaching Assistant

University of Glasgow

August 2024 – Ongoing   Glasgow, United Kingdom

Graduate Teaching Assistant for the courses **Introduction to Data Science and Systems**, **Machine Learning and Artificial Intelligence for Data Scientists**, and **Text as Data**.

University Lecturer

Hamedan University of Technology

April 2020 – February 2023   Hamedan, Iran

Lecturer of the courses **Machine Learning in Python**, **C/C++ Programming**, and **Signal and Systems**.

Researcher

Hamedan University of Technology

April 2020 – February 2023   Hamedan, Iran

Researcher in the areas of **Face Detection/Recognition algorithms on embedded systems**, **EEG-based AI decision-making systems**, and **medical image watermarking**.

SKILLS

- Proficient in modelling deep learning and machine learning frameworks for biomedical images/signals.
- Advanced Python and MATLAB development with a focus on object-oriented programming.
- Extensive expertise in machine learning frameworks such as PyTorch, Keras, and Scikit-learn.
- Proficient in Python packages for medical image analysis, such as MONAI, ITK, and NiBabel.
- Extensive experience in data manipulation (NumPy, Pandas).
- Academic manuscript preparation, technical writing, and LaTeX typesetting.

EDUCATION

Ph.D. in Computing Science

University of Glasgow

Sept 2023 – Ongoing

Thesis title: Bayesian Deep Atlases for Cardiac Motion Abnormality Assessment by Integrating Imaging and Metadata.

M.Sc. in Biomedical Engineering

Tarbiat Modares University

Sept 2016 – June 2019

Thesis title: Human sperm segmentation in microscopic images using deep learning techniques.

B.Sc. in Electronic Engineering

Buali-Sina University

Sept 2011 – June 2016

Thesis title: Study and research about ECU and vehicle electronic control systems.

PROJECTS

Anti-theft Vehicle System based on Face Recognition

Hamedan University of Technology

Developed a Raspberry Pi–based vehicle anti-theft prototype using face recognition.

A control System for face mask and body temperature monitoring

Hamedan University of Technology

Developed a Raspberry Pi–based mask detection and temperature monitoring system using CNN models.

Prediction of depression based on EEG Signals

Baqiyatallah University of Medical Sciences

Developed EEG-based machine learning and deep learning frameworks for diagnosing major depressive disorder.

REFEREES

Dr. Ali Gooya
University Of Glasgow
Ali.Gooya@glasgow.ac.uk

PUBLICATIONS

Journal Articles

- R. Akbari Movahed, A. Rezaee, A. Zakeri, C. Berry, E. S. L. Ho, and A. Gooya, "Cardiomorphnet: Cardiac motion prediction using a shape-guided bayesian recurrent deep network," *arXiv preprint arXiv:2508.20734*,, 2025, Submitted to Medical Image Analysis. arXiv: 2508.20734 `[cs.CV]`.
- A. Afzali, A. Khaleghi, B. Hatef, R. Akbari Movahed, and G. Pirzad Jahromi, "Automated major depressive disorder diagnosis using a dual-input deep learning model and image generation from eeg signals," *Waves in Random and Complex Media*, pp. 1–16, 2023.
- R. A. Movahed, G. P. Jahromi, S. Shahyad, and G. H. Meftahi, "A major depressive disorder diagnosis approach based on eeg signals using dictionary learning and functional connectivity features," *Physical and Engineering Sciences in Medicine*, pp. 1–15, 2022.
- R. A. Movahed and M. Rezaeian, "Automatic diagnosis of mild cognitive impairment based on spectral, functional connectivity, and nonlinear eeg-based features," *Computational and Mathematical Methods in Medicine*, vol. 2022, 2022.
- R. A. Movahed, G. P. Jahromi, S. Shahyad, and G. H. Meftahi, "A major depressive disorder classification framework based on eeg signals using statistical, spectral, wavelet, functional connectivity, and nonlinear analysis," *Journal of Neuroscience Methods*, vol. 358, p. 109 209, 2021.
- R. A. Movahed, E. Mohammadi, and M. Orooji, "Automatic segmentation of sperm's parts in microscopic images of human semen smears using concatenated learning approaches," *Computers in Biology and Medicine*, vol. 109, pp. 242–253, 2019.

Conference Proceedings

- R. A. Movahed, N. E. Hamedani, S. Z. Sadredini, and M.-r. Rezaeian, "An automated eeg-based mild cognitive impairment diagnosis framework using spectral and functional connectivity features," in *2021 28th National and 6th International Iranian Conference on Biomedical Engineering (ICBME)*, IEEE, 2021, pp. 271–275.
- R. A. Movahed, M. Rezaeian, S. Javadifar, and M. Alimoradijazi, "A face recognition framework based on the integration of eigenfaces algorithm and image registration technique," in *2020 27th National and 5th International Iranian Conference on Biomedical Engineering (ICBME)*, IEEE, 2020, pp. 26–30.
- R. A. Movahed, M. R. Rezaeian, and S. Ghasemi, "An image watermarking algorithm for medical computerized tomography images," in *2019 5th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS)*, IEEE, 2019, pp. 1–5.
- R. A. Movahed and M. Orooji, "A learning-based framework for the automatic segmentation of human sperm head, acrosome and nucleus," in *2018 25th National and 3rd International Iranian Conference on Biomedical Engineering (ICBME)*, IEEE, 2018, pp. 1–6.